Homework 3

MAT2040 : Summer 2023 : Prof. Lun Yu

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- Due date: 2023.6.30, 11.59 p.m.
- Late submission is NOT accepted.
- Please submit your answers as a PDF file with a name like "XXXXXXXXX HW3.pdf" (Your student ID + HW No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size $\leq$ 10MB).
- Please make sure that your submitted file is clear and readable. Submitted file that can not be opened or not readable will get 0 point.

Problem 1

Find the inverses of the matrices in (i) and (ii).

(i) $\begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & -2 & 1 \\ 4 & -7 & 3 \\ -2 & 6 & -4 \end{bmatrix}$

Answer

1.

(i)

$$(A \mid I) = \begin{pmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & -3 & 8 & -2 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 1 & \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{pmatrix} = \left(I \mid A^{-1}\right),$$

thus $A^{-1} = \begin{pmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{pmatrix}$.

(ii)

$$(A \mid I) = \begin{pmatrix} 1 & -2 & 1 & 1 & 0 & 0 \\ 4 & -7 & 3 & 0 & 1 & 0 \\ -2 & 6 & -4 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -4 & 1 & 0 \\ 0 & 2 & -2 & 2 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -4 & 1 & 0 \\ 0 & 0 & 0 & 10 & -2 & 1 \end{pmatrix}$$

thus A^{-1} does not exist.

Problem 2

(i) Find invertible matrices A and B such that $A + B$ is not invertible.

(ii) Find singular matrices A and B such that $A + B$ is invertible.

Answer

(i) Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, B = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$, then $A + B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

Thus A, B are invertible, but $A + B$ is not invertible.

(ii) Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, then $A + B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Thus A, B are singular, but $A + B$ is invertible.

Problem 3

Find the numbers a and b such that :

$$\begin{bmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{bmatrix}.$$

Answer

Since

$$\begin{pmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{pmatrix} \begin{pmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{pmatrix} = I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

we have

$$\begin{cases} 4a - 3b = 1 \\ 4b - a - 2b = 0 \end{cases} \Rightarrow \begin{cases} a = \frac{2}{5} \\ b = \frac{1}{5} \end{cases}.$$

Problem 4

Prove that A is invertible if $a \neq 0$ and $a \neq b$ (find the pivots or A^{-1}). Then find three numbers c so that C is not invertible:

$$A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix} \quad C = \begin{bmatrix} 2 & c & c \\ c & c & c \\ 8 & 7 & c \end{bmatrix}.$$

Answer

$$A = \begin{pmatrix} a & b & b \\ a & a & b \\ a & a & a \end{pmatrix} \Rightarrow \begin{pmatrix} a & b & b \\ 0 & a-b & 0 \\ 0 & a-b & a-b \end{pmatrix} \Rightarrow \begin{pmatrix} a & b & b \\ 0 & a-b & 0 \\ 0 & 0 & a-b \end{pmatrix}$$

Since $a \neq 0, a \neq b, A$ have 3 nonzero pivots, then we have A is invertible.

$$C = \begin{pmatrix} 2 & c & c \\ c & c & c \\ 8 & 7 & c \end{pmatrix} \Rightarrow \begin{pmatrix} 2 & c & c \\ 0 & c-\frac{c^2}{2} & c-\frac{c^2}{2} \\ 0 & 7-4c & -3c \end{pmatrix}.$$

If $c - \frac{c^2}{2} = 0, C$ cannot have 3 nonzero pivots.

Thus, if $c = 0, c = 2, C$ is not invertible. For $c - \frac{c^2}{2} \neq 0$:

$$\begin{pmatrix} 2 & c & c \\ 0 & c-\frac{c^2}{2} & c-\frac{c^2}{2} \\ 0 & 7-4c & -3c \end{pmatrix} \Rightarrow \begin{pmatrix} 2 & c & c \\ 0 & 1 & 1 \\ 0 & 7-4c & -3c \end{pmatrix} \Rightarrow \begin{pmatrix} 2 & c & c \\ 0 & 1 & 1 \\ 0 & 0 & c-7 \end{pmatrix}$$

If $c = 7, C$ is not invertible. Overall, if $c = 0, 2, 7, C$ is not invertible.

Problem 5

Let $A^N = 0$ for a certain integer $N > 0$, prove that $(I - A)^{-1} = I + A + A^2 + \cdots + A^{N-1}$.

Answer

$$\begin{aligned} (I-A)\left(I + A + A^2 + \cdots + A^{N-1}\right) &= I + A + \cdots + A^{A-1} - A - A^2 - \cdots - A^{N-1} - A^N \\ &= I - A^N \\ &= I, \end{aligned}$$

similarly, $\left(I + A + A^2 + \cdots + A^{N-1}\right)(I - A) = I$.
Thus $(I - A)^{-1} = I + A + A^2 + \cdots + A^{N-1}$.

Problem 6

(i) If $A = LDU$ and also $A = L_1 D_1 U_1$ with all factors invertible, prove that $L = L_1$, $D = D_1$ and $U = U_1$.

(ii) If $A = LU$ and also $A = L_1 U_1$ with all factors invertible, is $L = L_1$ and $U = U_1$ still true?

(iii) If $PA = LDU$ and also $P_1 A = L_1 D_1 U_1$ with all factors invertible, is $L = L_1$, $U = U_1$, $D = D_1$ and $P = P_1$ true?

Answer

(i) $A = LDU = L_1D_1U_1$, then we have $L_1^{-1}LD = D_1U_1U^{-1}$.

Recall, the inverse of lower-triangular matrix is still lower-triangular matrix, and the product of lower-triangular matrices is still lower-triangular matrix.

Thus, $L_1^{-1}LD = D_1U_1U^{-1}$ is both lower-triangular matrix and upper-triangular matrix, so $L^{-1}LD = D_1U_1U^{-1}$ is a diagonal matrix.

So we have $L_1^{-1}L = I, U_1U^{-1} = I$ and $D = D_1$, which means $L = L_1$ and $U = U_1$.

(ii) Let $A = \bar{L}\bar{D}\bar{U}$, by (i), we know $\bar{L}, \bar{D}, \bar{U}$ is unique.

$A = LU = L_1U_1$, we can let $L = \bar{L}, U = \bar{D}\bar{U}$ and $L_1 = \bar{L}\bar{D}$, $U_1 = \bar{U}$.

Since $\bar{D} = I$ is not hold for all A, $L = L_1$ and $U = U_1$ is not true.

(iii) Not true, for example, consider $A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$, we can let

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P_1 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Problem 7

Tridiagonal matrices have zero entries except on the main diagonal and the two adjacent diagonals. Factor these into $A = LU$ and $A = LDL^{\mathrm{T}}$:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} a & a & 0 \\ a & a+b & b \\ 0 & b & b+c \end{bmatrix}.$$

Answer

For $A = \begin{pmatrix} a & a & 0 \\ a & a+b & b \\ 0 & b & b+c \end{pmatrix}$,

let $L_1 = \begin{pmatrix} 1 & & \\ -1 & 1 & \\ & & 1 \end{pmatrix}$, then $L_1A = \begin{pmatrix} a & a & 0 \\ 0 & b & b \\ 0 & b & b+c \end{pmatrix}$.

Let $L_2 = \begin{pmatrix} 1 & & \\ & 1 & \\ & -1 & 1 \end{pmatrix}$, then $L_2L_1A = \begin{pmatrix} a & a & 0 \\ 0 & b & b \\ 0 & 0 & c \end{pmatrix} = DU$.

$L_2L_1A = DU \Rightarrow A = L_1^{-1}L_2^{-1}DU$.Thus

$$L = L_1^{-1}L_2^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, D = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}, U = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

For $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$, means $a = 1, b = 1, c = 1$.

Thus $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} = LDU = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

Problem 8

Theorem of multiplicative property: If A and B are $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$. Please verify this theorem for $A = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$.

Answer

$|A| = \begin{vmatrix} 6 & 1 \\ 3 & 2 \end{vmatrix} = 12 - 3 = 9,$

$|B| = \begin{vmatrix} 4 & 3 \\ 1 & 2 \end{vmatrix} = 8 - 3 = 5,$

$AB = \begin{pmatrix} 6 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 25 & 20 \\ 14 & 13 \end{pmatrix}, |AB| = 25 \times 13 - 20 \times 14 = 45.$

Thus $\det(AB) = \det(A)\det(B)$.

Problem 9

For each of the following, compute (i) $\det(A)$, and (ii)A^{-1}.

(i) $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ (ii) $A = \begin{bmatrix} 3 & 1 \\ 2 & 6 \end{bmatrix}$ (iii) $A = \begin{bmatrix} 3 & 3 & 1 \\ 2 & 1 & 1 \\ -2 & 2 & -1 \end{bmatrix}$ (iv) $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Answer

(i) $det(A) = -5$, $A^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{1}{5} & -\frac{1}{5} \end{bmatrix}$

(ii) $det(A) = 16$, $A^{-1} = \begin{bmatrix} \frac{3}{8} & -\frac{1}{16} \\ -\frac{1}{8} & \frac{3}{16} \end{bmatrix}$

(iii) $det(A) = -3$, $A^{-1} = \begin{bmatrix} 1 & -\frac{5}{3} & -\frac{2}{3} \\ 0 & \frac{1}{3} & \frac{1}{3} \\ -2 & 4 & 1 \end{bmatrix}$

(iv) $det(A) = 2$, $A^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$

Problem 10

Compute the determinants of the following matrices.
(i)

$$A = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 7 & 2 & 1 & -3 \end{bmatrix}$$

(ii)

$$A = \begin{bmatrix} 7 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -6 & 5 & 7 & 2 \\ 2 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{bmatrix}$$

(iii)

$$A = \begin{bmatrix} 3 & 7 & 8 & 9 & -5 \\ 0 & 2 & -7 & 3 & 9 \\ 0 & 0 & 2 & 8 & 1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

(iv)

$$A = \begin{bmatrix} 8 & 2 & 1 & 0 & 2 \\ 2 & 0 & 1 & 3 & 4 \\ 1 & 5 & 2 & 2 & 0 \\ 0 & 4 & 9 & 2 & 0 \end{bmatrix}$$

Answer

(i) 36; (ii) 12; (iii) non-existent; (iv) non-existent;

Problem 11

Let A and B be 3×3 matrices, with det(A) = 3 and det(B) = 4. Use properties of determinants to compute:
(i) $det(AB)$
(ii) $det(5A)$
(iii) $det(B^T)$
(iv) $det(A^{-1})$
(v) $det(A^3)$

Answer

(i) $det(AB) = det(A)det(B) = 12$
(ii) $det(5A) = 5^3 det(A) = 375$
(iii) $det(B^T) = det(B) = 4$
(iv) $det(A^{-1}) = \dfrac{1}{det(A)} = \dfrac{1}{3}$
(v) $det(A^3) = (det(A))^3 = 27$

Problem 12

Let A be a $k \times k$ matrix and let B be an $(n-k)\times(n-k)$ matrix. Let $E = \begin{bmatrix} I_k & O \\ O & B \end{bmatrix}$, $F = \begin{bmatrix} A & O \\ O & I_{n-k} \end{bmatrix}$, $C = \begin{bmatrix} A & O \\ O & B \end{bmatrix}$ where I_k and I_{n-k} are the $k \times k$ and $(n-k) \times (n-k)$ identity matrices.
(i) Show that $det(E) = det(B)$.
(ii) Show that $det(F) = det(A)$.
(iii) Show that $det(C) = det(A)det(B)$.

Answer

(i) Expanding $det(E)$ by cofactors along the first row, we have

$$det(E) = 1 \cdot det(E_{11})$$

Similarly expanding $det(E_{11})$ along the first row yields

$$det(E_{11}) = 1 \cdot det((E_{11})_{11})$$

So after k steps of expanding the submatrices along the first row, we get

$$det(E) = 1 \cdot 1 \cdot 1 \cdots det(B) = det(B)$$

(ii) The argument here is similiar to that in part (i) except that at each step we expand along the last row of the matrix. After $n-k$ steps, we get

$$det(F) = 1 \cdot 1 \cdots det(A) = det(A)$$

(iii) Since $C = EF$, it folows that

$$det(C) = det(E)det(F) = det(B)det(A) = det(A)det(B)$$

Problem 13

Compute the determinants of the following matrices.
(i)

$$\begin{bmatrix} a & 1 & 0 & 0 \\ -1 & b & 1 & 0 \\ 0 & -1 & c & 1 \\ 0 & 0 & -1 & d \end{bmatrix}$$

(ii)

$$\begin{bmatrix} a & b & b & \dots & b \\ b & a & b & \dots & b \\ b & b & a & \dots & b \\ \dots & \dots & \dots & \dots & \dots \\ b & b & b & \dots & a \end{bmatrix}$$

(iii)

$$\begin{bmatrix} \alpha+\beta & \alpha\beta & 0 & 0 & \dots & 0 \\ 1 & \alpha+\beta & \alpha\beta & \cdots & 0 & 0 \\ 0 & 1 & \alpha+\beta & \cdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \alpha+\beta & \alpha\beta \\ 0 & 0 & 0 & \dots & 1 & \alpha+\beta \end{bmatrix}$$

Answer

(i)

$$\begin{vmatrix} a & 1 & 0 & 0 \\ -1 & b & 1 & 0 \\ 0 & -1 & c & 1 \\ 0 & 0 & -1 & d \end{vmatrix} \xrightarrow[r_3 \leftarrow r_3 + \frac{r_2}{b+\frac{1}{a}};\ r_4 \leftarrow r_4 + \frac{r_3}{c+\frac{1}{b+\frac{1}{a}}}]{r_2 \leftarrow r_2 + \frac{r_1}{a}} \begin{vmatrix} a & 1 & 0 & 0 \\ 0 & b+\frac{1}{a} & 1 & 0 \\ 0 & 0 & c+\frac{1}{b+\frac{1}{a}} & 1 \\ 0 & 0 & 0 & d+\frac{1}{c+\frac{1}{b+\frac{1}{a}}} \end{vmatrix}$$

$$= abcd + ab + ad + cd + 1$$

(ii)

$$\begin{vmatrix} a & b & b & \dots & b \\ b & a & b & \dots & b \\ b & b & a & \dots & b \\ \dots & \dots & \dots & \dots & \dots \\ b & b & b & \dots & a \end{vmatrix} \xrightarrow{c_1 \leftarrow \sum_{i=1}^{n} c_i} \begin{vmatrix} a+(n-1)b & b & b & \dots & b \\ a+(n-1)b & a & b & \dots & b \\ a+(n-1)b & b & a & \dots & b \\ \dots & & \dots & \dots & \dots & \dots \\ a+(n-1)b & b & b & \dots & a \end{vmatrix}$$

$$\xrightarrow[\cdots r_n \leftarrow r_n - r_1]{r_2 \leftarrow r_2 - r_1;\ r_3 \leftarrow r_3 - r_1} \begin{vmatrix} a+(n-1)b & b & b & \dots & b \\ 0 & a-b & 0 & \dots & 0 \\ 0 & 0 & a-b & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & a-b \end{vmatrix}$$

$$= [a+(n-1)b]\,(a-b)^{n-1}$$

(iii) Let $d_n = \begin{vmatrix} \alpha+\beta & \alpha\beta & 0 & 0 & \dots & 0 \\ 1 & \alpha+\beta & \alpha\beta & \cdots & 0 & 0 \\ 0 & 1 & \alpha+\beta & \cdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \alpha+\beta & \alpha\beta \\ 0 & 0 & 0 & \dots & 1 & \alpha+\beta \end{vmatrix}$

$$d_n = (\alpha+\beta)d_{n-1} - \alpha\beta \begin{vmatrix} 1 & \alpha\beta & 0 & \cdots & 0 & 0 \\ 0 & \alpha+\beta & \alpha\beta & \cdots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \alpha+\beta & \alpha\beta \\ 0 & 0 & 0 & \dots & 1 & \alpha+\beta \end{vmatrix}_{n-1} = (\alpha+\beta)d_{n-1} - \alpha\beta d_{n-2}$$

$d_n - \alpha d_{n-1} = \beta(d_{n-1} - \alpha d_{n-2}) = \beta^2(d_{n-2} - \alpha d_{n-3}) = \beta^{n-2}(d_2 - \alpha d_1)$
$d_2 = (\alpha+\beta)^2 - \alpha\beta$
$d_1 = \alpha + \beta$
$d_n - \alpha d_{n-1} = \beta^n$
$d_{n-1} - \alpha d_{n-2} = \beta^{n-1}$
$\cdots$
$d_2 - \alpha d_1 = \beta^2$
$d_n - \alpha^{n-1} d_1 = \beta^n + \alpha\beta^{n-1} + \alpha^2\beta^{n-2} + \cdots + \alpha^{n-2}\beta^2$
$d_n = \beta^n + \alpha\beta^{n-1} + \alpha^2\beta^{n-2} + \cdots + \alpha^{n-2}\beta^2 + \alpha^{n-1}\beta + \alpha^n$

Problem 14

Let A and B be 2×2matrices

$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ $C = \begin{bmatrix} a_{11} & a_{12} \\ b_{21} & b_{22} \end{bmatrix}$ $D = \begin{bmatrix} b_{11} & b_{12} \\ a_{21} & a_{22} \end{bmatrix}$
Show that $det(A+B) = det(A) + det(B) + det(C) + det(D)$.

Answer

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix}$$

$det(A+B)$
$= (a_{11} + b_{11})(a_{22} + b_{22}) - (a_{12} + b_{12})(a_{21} + b_{21})$
$= a_{11}a_{22} + a_{11}b_{22} + b_{11}a_{22} + b_{11}b_{22} - a_{21}a_{12} - a_{21}b_{12} - b_{21}a_{12} - b_{21}b_{12}$
$= (a_{11}a_{22} - a_{21}a_{12}) + (b_{11}b_{22} - b_{21}b_{12}) + (a_{11}b_{22} - b_{21}a_{12}) + (b_{11}a_{22} - a_{21}b_{12})$
$= det(A) + det(B) + det(C) + det(D)$

Problem 15

Let U be a square matrix such that $U^TU = I$. Show that $det(U) = \pm 1$.

Answer

$det(U^TU) = (det(U))^2 = 1$. Hence, $det(U) = \pm 1$.

Problem 16

Let a, b, c, $d \in \mathbb{R}$ and

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^3 & b^3 & c^3 & d^3 \end{bmatrix}$$

(i) Show that $det(A) = (d-a)(c-a)(b-a)(d-b)(c-b)(d-c)$.
(ii) Show a, b, c, d are distinct if and only if A is nonsingular.

Answer

(i) Operate $-aR_3 + R_4, -aR_2 + R_3, -aR_1 + R_2$ successively and expand along the first column, we get

$$det(A) = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & b-a & c-a & d-a \\ 0 & b(b-a) & c(c-a) & d(d-a) \\ 0 & b^2(b-a) & c^2(c-a) & d^2(d-a) \end{vmatrix}$$

$$= (b-a)(c-a)(d-a) \begin{vmatrix} 1 & 1 & 1 \\ b & c & d \\ b^2 & c^2 & d^2 \end{vmatrix}$$

Operate $-bR_2 + R_3, -bR_1 + R_2$ and expand along the first column, we get

$$\begin{vmatrix} 1 & 1 & 1 \\ b & c & d \\ b^2 & c^2 & d^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & c-b & d-b \\ 0 & c(c-b) & d(d-b) \end{vmatrix} = (c-b)(d-b) \begin{vmatrix} 1 & 1 \\ c & d \end{vmatrix} = (c-b)(d-b)(d-c)$$

Thus, $det(A) = (d-a)(c-a)(b-a)(d-b)(c-b)(d-c)$
(ii) From the result of (i), we can see that a, b, c, d are distinct if and only if $det(A) \neq 0$. In other words, a, b, c, d are distinct if and only if A is nonsingular.